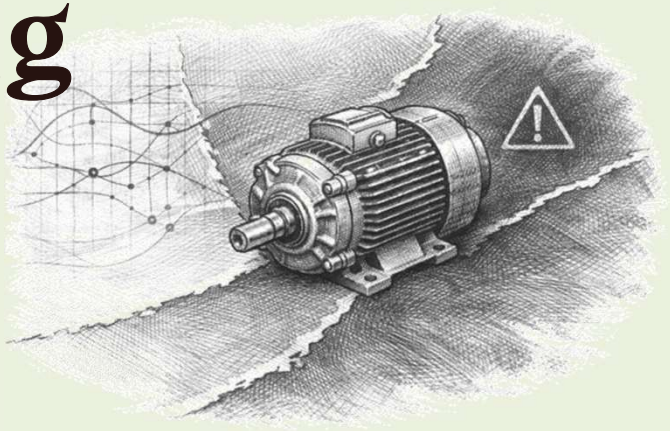
A practical approach to predictive maintenance

Motor Failures Prediction using KNN.



Probability Modeling | Risk Estimation | Scenario Simulation

The problem: Early Failure Detection Is Not Binary

- Electric motors rarely fail suddenly.
- Degradation develops gradually through vibration, temperature, load, and electrical behavior.
- Individual signals may remain within acceptable limits.
- Risk emerges when **multiple conditions interact**, not when a single threshold is crossed.

The challenge is not detecting failure — it’s detecting incipient risk under uncertainty.

Data is never perfect

- Motor condition data comes from **multiple sources**, not a single system.
- Each signal reflects a different measurement process with its own noise and limitations.
- Sensor readings are affected by calibration, environment, and operating context.
- Failure labels are often **imperfect or delayed**, especially in early-stage degradation.

Predictive models must operate under uncertainty — not ideal conditions.

Why KNN for this problem?

- Failure risk does not follow a simple linear trend.
- Similar operating conditions tend to lead to similar outcomes.
- KNN evaluates risk based on **proximity to historical operating states.**
- The model makes no assumptions about functional form or distribution.

Risk emerges in neighborhoods, not equations.

Signals and Data Sources

- The model uses condition-monitoring signals commonly available in industrial environments.
- Each variable reflects a different physical aspect of motor behavior.

Key inputs include

- Vibration (RMS, peak-to-peak, dominant frequency) → Condition monitoring systems, accelerometers, vibration routes.
- Bearing temperature → Embedded sensors, infrared inspections.
- Electrical current → Motor control centers (MCC), power analyzers.
- Load percentage → PLC signals, production context.

Target variable

- Incipient failure

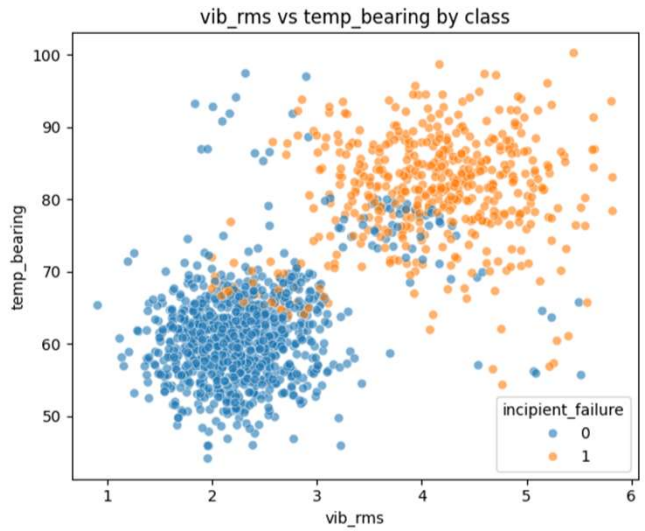
  0 → Normal operation

  1 → Early-stage failure condition

Each signal alone is incomplete — value emerges when they are analyzed together.

Exploratory Insight: Structure Exists Despite Noise

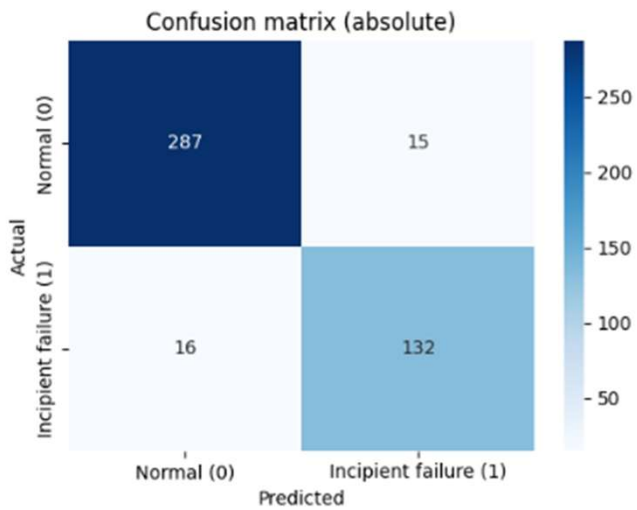
- Even with noisy measurements, structure is visible in the data.
- Normal and failure conditions tend to occupy different regions.
- Overlap exists in intermediate operating conditions.
- This overlap represents **early degradation and uncertainty**, not error.

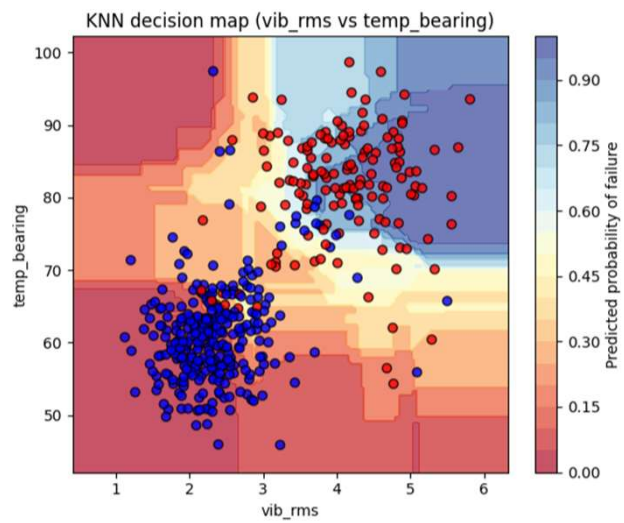


Separation is imperfect — and that's exactly the point.

Decision Boundary: Where Risk Emerges

- The model separates operating regions based on similarity to historical conditions.
- Clear safe and high-risk regions emerge in the feature space.
- A gray zone appears where uncertainty is highest.
- Most misclassifications occur near this boundary.





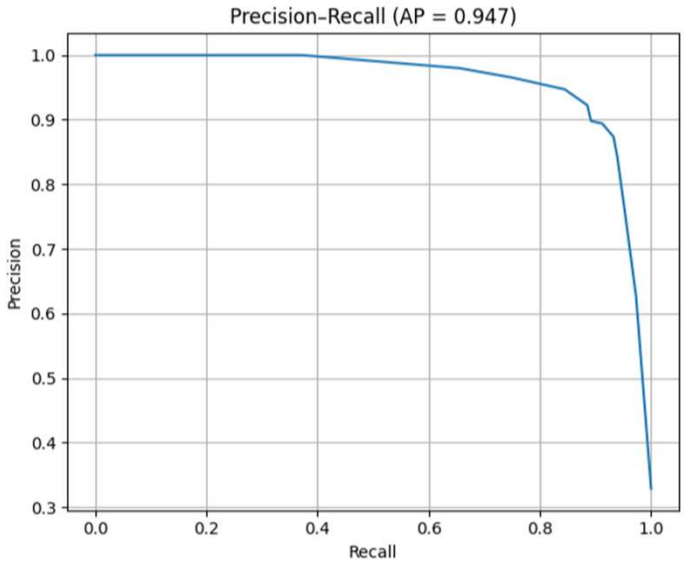
The gray zone is where engineering judgment and decision-making matter most.

Model Performance: Honest Metrics

- Performance is evaluated on unseen test data.
- Metrics reflect realistic conditions with noise and class overlap.
- Accuracy and recall remain strong without reaching artificial perfection.
- Errors concentrate in borderline operating conditions.

Key metrics

- Accuracy: ~0.93–0.95
- ROC–AUC: ~0.95
- Recall (incipient failure): prioritized over precision



The model is reliable enough to support decisions — not to replace engineering judgment.

Comparison with Baseline Models

- Logistic Regression provides a clear and interpretable linear baseline.
- Naive Bayes offers a fast probabilistic reference.
- Both baselines struggle to capture localized, non-linear patterns.
- KNN consistently performs better under overlapping and noisy conditions.

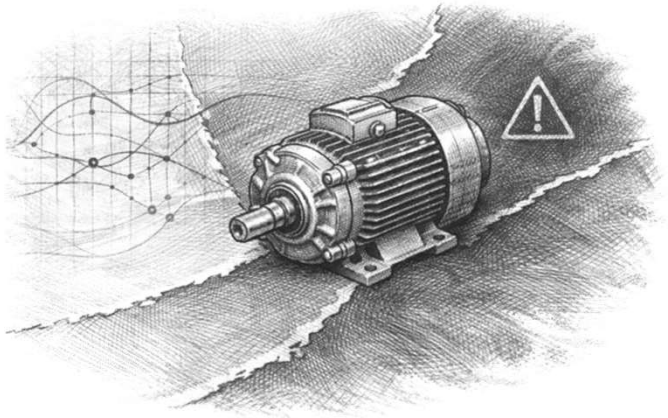
	Model	Accuracy	ROC-AUC	F1 (failure=1)
0	KNN	0.931111	0.968789	0.894915
1	Logistic Regression	0.928889	0.979484	0.892617
2	Naive Bayes	0.942222	0.968118	0.913333

Model complexity is justified by problem structure, not by metrics alone.

Final reflection

Electric motors rarely fail because of a single variable drifting out of range.
Failure emerges when vibration, temperature, load, and electrical behavior interact over time.

K-Nearest Neighbors allows us to reason in terms of **similar operating conditions**, rather than fixed equations or rigid thresholds.
It answers a simple but powerful question:



It is a conversation:
Which operating conditions are we accepting today?
Where is degradation quietly accumulating?
Which gray zones deserve attention before alarms appear?

Predictive maintenance is not about replacing engineering judgment.
It is about supporting it with evidence.

Take it further

Feel free to download the full project:
the dataset, the notebook, and the simulator.

Change the data.
Break the code.
Ask different questions.

This model is just one snapshot in an ongoing process.
And that process never truly stops.

Because improvement has no finish line.